

Dr Waseem Raza, PhD

Machine Learning Engineer | Generative & Agentic AI

LLM Applications | RAG | Agent Orchestration | Cloud Data Pipelines

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Professional Summary

AI/ML and Generative AI Engineer with a PhD in Electrical and Computer Engineering and 7+ years of experience developing data-driven systems across enterprise, industry, and research environments. Experience includes LLM-enabled analytics, AI-ready data pipelines, domain-adaptive machine learning, and scalable cloud solutions using Python, SQL, Azure Databricks, Apache Spark, and Azure services. At AT&T, lead development and operational integration of AI solutions for critical service-assurance workflows, improving automated diagnostic accuracy from 20% to 80%. Combine model evaluation, data quality, ETL, and cross-functional delivery to translate technical requirements into production-oriented AI capabilities.

Technical Expertise, Leadership, and Professional Strengths

Generative & Agentic AI: LLM applications, retrieval-augmented generation (RAG), prompt engineering, document ingestion, chunking, embeddings, semantic retrieval, structured information extraction, tool-using agents, agent orchestration, and grounded response generation

Microsoft AI Ecosystem: Microsoft Foundry, Azure OpenAI, Azure AI Services, Azure AI Search, Azure Machine Learning, Foundry models and agents, Azure Content Understanding, Semantic Kernel, and responsible AI concepts

LLM Frameworks & Models: Hugging Face Transformers, LangChain, LangGraph, OpenAI APIs, transformer architectures, fine-tuning concepts, NLP foundation models, prompt templates, structured outputs, and LLM evaluation

Data & AI Platforms: Azure Databricks, Apache Spark, PySpark, Azure Data Factory, Azure Synapse Analytics, Azure Functions, Azure Blob Storage, Azure Data Lake, Google BigQuery, and cloud analytics workflows

Data Engineering: ETL/ELT pipelines, AI-ready datasets, batch processing, ingestion, cleansing, schema transformation, validation, feature engineering, data-quality assessment, and structured and unstructured data workflows

Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, transfer learning, domain adaptation, adversarial learning, anomaly detection, classification, regression, time-series forecasting, cross-validation, and model evaluation

MLOps & Software Engineering: Docker, Kubernetes, OpenShift, Git, GitHub Actions, CI/CD, REST APIs, Flask, testing, Linux, shell scripting, model deployment, reproducible workflows, and cloud-native application development

Programming & Analytics: Python, SQL, C/C++, and R; Pandas, NumPy, SciPy, Statsmodels, Matplotlib, Plotly, Streamlit, Jupyter, and exploratory data analysis

Leadership and Professional Strengths

- **Ownership and accountability:** Lead AI/ML initiatives through requirements analysis, data preparation, model development, validation, stakeholder review, and deployment planning.
- **Technical leadership:** Define technical approaches, evaluate alternatives, coordinate experimentation, and communicate findings to guide projects from ambiguous requirements to measurable outcomes.
- **Cross-functional collaboration:** Work with software engineers, network specialists, researchers, geologists, and public-safety stakeholders to translate domain requirements into practical data and AI solutions.
- **Continuous learning:** Apply ongoing study in generative AI, agentic AI, RAG, LLM evaluation, and cloud-native platforms through hands-on projects and professional coursework.
- **Reliability and commitment:** Maintain reproducible workflows, careful validation, and attention to detail throughout experimentation and delivery.
- **Adaptability and problem solving:** Combine critical thinking, systematic experimentation, and data-driven analysis to address unfamiliar technologies and complex operational problems.

Professional Experience

Machine Learning Engineer

AT&T, contractor via Innovatix Inc.

Feb. 2025-Present

Bedminster, NJ

- Own end-to-end development, testing, and operational integration of an AI-driven diagnostic system for critical E911 service-assurance workflows, improving automated root-cause classification accuracy from 20% to 80%.
- Designed domain-adaptive neural models and reusable data-processing workflows that generalize across heterogeneous operational domains and reduce performance degradation caused by distribution shifts.
- Developed multimodal AI prototypes combining work-order text, location information, network context, and operational signals to support GenAI-assisted diagnostics for VoNR and 5G service assurance.
- Created model-validation, error-analysis, and failure-mode assessment procedures to compare performance across domains and support reliable deployment in high-impact operational environments.
- Partnered with network engineers, software teams, and public-safety stakeholders to translate operational requirements into validated AI workflows and production-oriented data assets.

Data Scientist and AI/ML Engineer

Impac Exploration Services

Sep. 2024-Dec. 2024

Weatherford, OK

- Built an LLM-enabled analytical pipeline for extracting and interpreting information from geochemical well data.
- Developed scalable ingestion, cleansing, transformation, and validation workflows using Python, Azure Databricks, Azure Functions, and Synapse Analytics.
- Automated ETL processing for large exploration datasets, reducing data-preparation time by 40% and improving reproducibility.
- Implemented cross-validation, data-quality, and consistency checks that improved reliability of analytical outputs by 20%.

- Collaborated with geologists to convert domain requirements and technical information into AI-ready datasets and actionable analytical outputs.

Advanced-Degrees Intern
AT&T Labs

Jun. 2024-Aug. 2024
Austin, TX

- Analyzed 500K+ LTE/5G-NR diagnostic logs and applied ML methods to detect anomalies, reducing diagnostic time by 20%.
- Investigated 5G NR-NTN cell-selection behavior and improved decision algorithms by 15% through data-driven analysis and validation.
- Conducted structured analysis of large operational datasets and communicated technical findings to engineering teams supporting next-generation wireless systems.

Advanced Wireless Technology Development Intern
Futurewei Technologies

Sep. 2022-Jan. 2023
Schaumburg, IL

- Improved autoencoder-based channel-state-information compression by 25% using transfer learning and systematic experimentation on large wireless datasets.
- Conducted 50+ experiments to validate model performance, robustness, and generalization under changing data conditions.
- Collaborated with research and engineering stakeholders to translate modeling results into recommendations for next-generation communication systems.

Advanced-Degrees Intern
AT&T Labs

Jun. 2022-Aug. 2022
Bedminster, NJ

- Designed ML and statistical methods for analyzing users’ timing in 5G standalone networks, improving analytical precision by 20%.
- Analyzed large-scale user trace data with SQL and visualization tools and communicated findings to senior technical leadership.
- Converted raw operational traces into interpretable performance indicators to support diagnosis and network optimization.

Graduate Research Assistant
AI4Networks, University of Oklahoma

Sep. 2019-Sep. 2024
Tulsa, OK

- Designed reusable Python simulation and data-generation platforms for large-scale experimentation, including a 3GPP-compliant network simulator and an AI-enabled 5G testbed.
- Built transfer-learning, adversarial-learning, GAN, autoencoder, and reinforcement-learning pipelines for sparse-data and cross-domain modeling.
- Developed robustness evaluation frameworks for models operating under distribution shifts, missing data, and limited labeled samples; led 100+ experiments from problem formulation through validation.
- Developed scalable self-healing and outage-management frameworks that improved modeled network resilience by 30%.
- Published peer-reviewed research, presented technical results internationally, and mentored junior researchers in ML implementation, experimental design, and technical writing.

Education

PhD, Electrical and Computer Engineering	09/2019-07/2025
University of Oklahoma, Norman, Oklahoma, USA	
MSc, Telecommunication Engineering	08/2014-08/2016
University of Engineering and Technology, Taxila, Pakistan	
BSc, Telecommunication Engineering	11/2010-06/2014
University of Engineering and Technology, Taxila, Pakistan	

Selected Projects

Agentic AI and RAG Applications
Portfolio Project

2026 | *In progress*

- Practical experiments with retrieval, tool-using agents, orchestration, memory, evaluation, and multi-agent workflows.

Tools & methods: LangGraph, CrewAI, AutoGen, RAG, Vector Databases

Automated Meeting Minutes Generator with LLMs
Guided Project

2025

- A workflow that transcribes meetings, produces structured summaries, and extracts decisions and action items for downstream collaboration.

Tools & methods: Python, Speech-to-Text, LLM Summarization, Workflow Automation

LLM-Based Extraction from Unstructured Documents
Guided Project

2025

- Document intelligence pipeline for extracting entities, fields, and tables from reports, invoices, and contracts.

Tools & methods: LLMs, OCR, Named Entity Recognition, Data Integration

AI-Ready Lakehouse Pipeline
Portfolio Project

2024

- An Azure Databricks data pipeline designed for analytics and downstream machine-learning workloads.

Tools & methods: Azure, Databricks, Spark, Data Engineering

Guided Project

- Documented architecture studies that translate product requirements into scalable services, data models, APIs, caching strategies, and reliability tradeoffs.

Tools & methods: System Design, APIs, Distributed Systems, Scalability

Certifications and Continuing Education

1. AWS Generative AI and AI Agents with Amazon Bedrock — [AWS/Coursera](#) | Completed (3/3) courses | 08/2026 | [Certificate](#)

Focus: Three-course professional certificate focused on developing, customizing, optimizing, and automating generative AI applications and intelligent agents with Amazon Bedrock.

Tools & Skills: Amazon Bedrock, Generative AI, AI Agents, Foundation Models, Knowledge Bases, Retrieval-Augmented Generation, LangChain, Amazon Q Developer, Model Customization, Prompt Engineering, Natural Language Processing, Text Generation, Summarization, Generative AI Application Development

2. IBM Generative AI Engineering Professional Certificate — IBM / Coursera | Completed (13/16) courses

Focus: Sixteen-course professional certificate developing job-ready skills across AI foundations, Python application development, machine learning, deep learning, NLP, transformer architectures, LLM fine-tuning, RAG, LangChain, and AI agents.

Tools & Skills: Generative AI Engineering, Large Language Models, Transformer Architectures, Prompt Engineering, RAG, LangChain, AI Agents, Hugging Face, PyTorch, Keras, Scikit-learn, Python, Flask, Gradio, PEFT, LoRA, QLoRA, RLHF, DPO, PPO, NLP, Machine Learning, Deep Learning, Vector Databases

3. IBM RAG and Agentic AI Professional Certificate — [IBM / Coursera](#) | Completed (8/8) courses | February 2026 | [Certificate](#)

Original eight-course credential.

Focus: Completed eight-course professional certificate covering retrieval-augmented generation, vector databases, multimodal applications, and agentic AI workflows.

Tools & Skills: Retrieval-Augmented Generation, Agentic AI, LangChain, LangGraph, Vector Databases, ChromaDB, FAISS, LlamaIndex, Multimodal AI, Tool Calling, Prompt Engineering, CrewAI, AutoGen / AG2, BeeAI, Gradio, Python, Flask

4. Generative AI Fundamentals Specialization — [IBM / Coursera](#) | Completed (5/5) courses | July 2026 | [Certificate](#)

Focus: Five-course foundation in generative AI, prompt engineering, LLMs, and practical enterprise applications.

Tools & Skills: Generative AI, Prompt Engineering, Foundation Models, Responsible AI, IBM watsonx, Hugging Face, Business Applications